

Time: 90 Minutes

Instructions: 1. Attempt all questions

2. Write your answers on your own paper

3. All necessary working must be shown

4. Marks will be deducted for careless or badly arranged work

QUESTION 1 (22 marks)

(a) Sketch each of the following curves on separate number planes

(i) $y = x^2 - 4$

2

(ii) $y = |x + 1|$

2

(iii) $x^2 + y^2 = 4$

2

(iv) $y = \frac{1}{x+2}$

2

(b) State whether each of the above curves are functions?

4

(c) State domain and range of

(i) $y = 2^{-x}$ (ii) $y = |x + 1|$ (iii) $y = \frac{1}{x+2}$ (iv) $x^2 + y^2 = 4$

8

(d) State the domain of $y = \frac{1}{\sqrt{8-2x}}$

2

QUESTION 2 (17 marks)(a) If $g(x) = x^2 - 4x$ find

(i) $g(2)$

(ii) $g(1-w)$

(iii) Find x if $g(x) = -3$

6

(b) Given that $f(x) = \begin{cases} 2x-4 & \text{if } x \geq 1 \\ x-3 & \text{if } -1 < x < 1 \\ -4x^2 & \text{if } x \leq -1 \end{cases}$

Find (i) $f(2)$

(ii) $f(0)$

(iii) $f(-1)$

(iv) Sketch the function

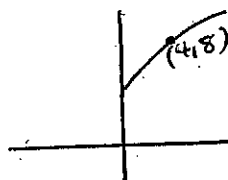
1

1

1

3

(c)

 $y=f(x)$ is an even function

(i) Redraw this function and complete the graph

2

(ii) Find $f(4) + f(-4)$

1

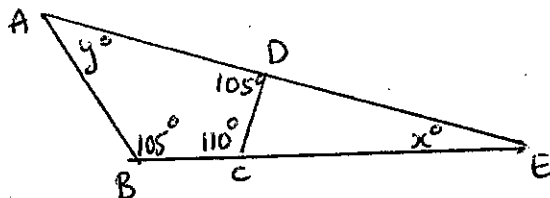
(d) Prove that $f(x) = x^3 - 4x$ is an odd function

2

QUESTION 3 (13 marks)

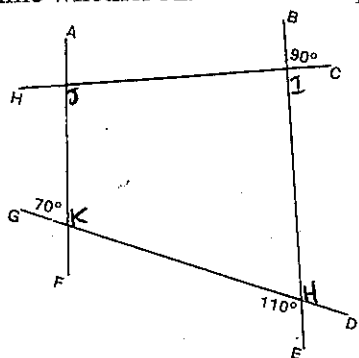
(a) Find the size of each interior angle of a regular octagon

2

(b) Find x and y giving reasons

4

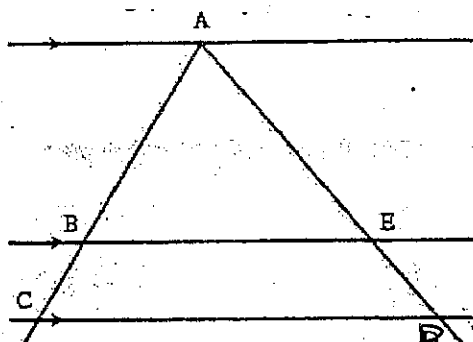
(c) Determine whether AF and BE are parallel with reasons



3

(d)

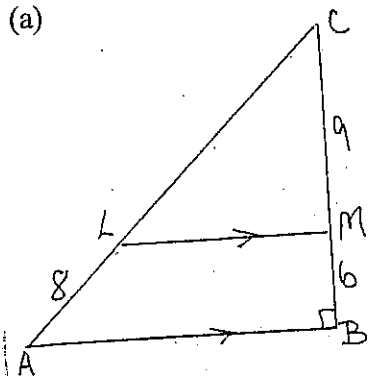
$AB=8\text{m}$, $BC=3\text{m}$ $ED=5\text{m}$
Find AD giving reasons



4

QUESTION 4 (11 marks)

(a)



(i) Find LC with reasons

2

(ii) Hence find AB as a surd

2

(b) A triangle has sides 12cm, 35cm, 37cm. Prove it is a right angled triangle.

2

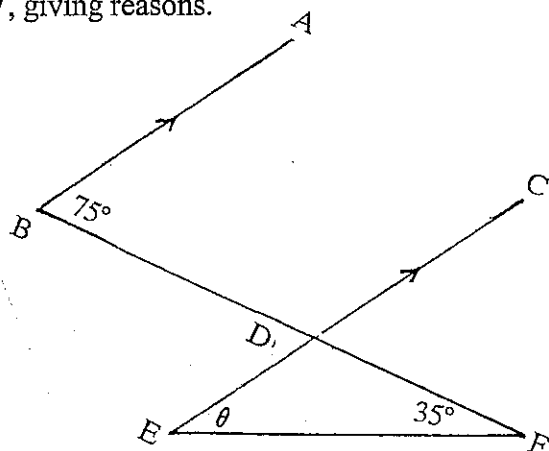
(c) A rhombus has perimeter 136cm and one diagonal is 60cm.

(i) Find the other diagonal as a surd

3

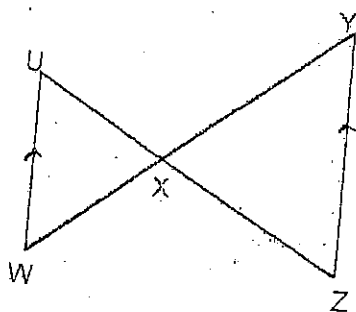
(ii) Find the area of the rhombus

2

QUESTION 5 (17 marks)(a) Find the size of θ , giving reasons.

4

(b) The lines UZ and WY intersect at X as in the figure, UW is 10cm, WX is 8cm, UX is 7cm and XZ is 12cm.



(i) Draw a neat sketch and mark it with all the given information.

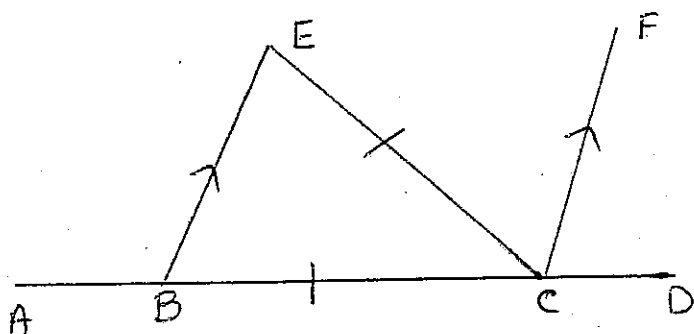
1

(ii) Prove that triangles UWX and XYZ are similar.

3

(iii) Hence or otherwise find the length of ZY with reasons

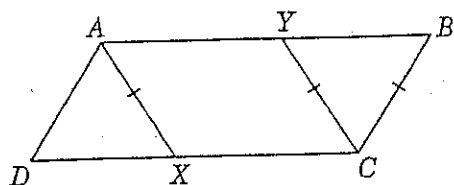
3

(c) ABCD is a straight line. Prove that CF bisects $\angle ECD$ giving clear reasons

6

QUESTION 6 (20 marks)

(a) ABCD is a parallelogram. The point X lies on CD the point Y lies on AB and $AX=CY=BC$ as shown in the diagram



(i) Copy the diagram onto your answer page

(ii) Explain why $\angle ADX = \angle CBY$

1

(iii) Show that $AD=AX$

3

(iv) Show that triangles ADX and CBY are congruent

4

(v) Hence prove that AYCX is a parallelogram

4

(b) ABCD is a rhombus, AX is perpendicular to BC and intersects BD at L

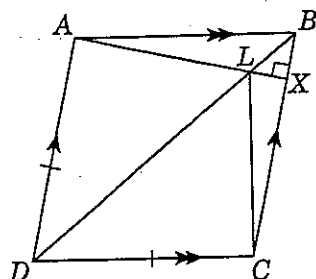


Figure not
to scale

(i) Copy the diagram onto your answer paper and state why

$$\angle ADB = \angle CDB$$

1

(ii) Prove that the triangles ALD and CLD are congruent

4

(iii) Show that $\angle DAL$ is a right angle with reasons

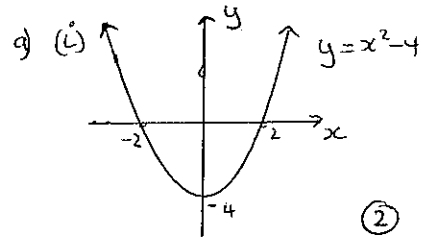
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(iv) Hence or otherwise find the size of $\angle LCD$ with reasons

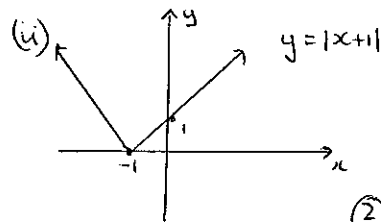
2

Year 11 Mathematics Task 3

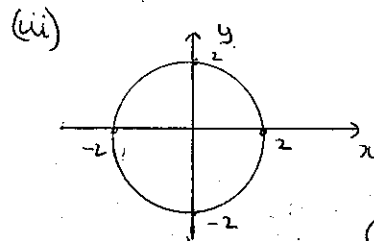
Question 1



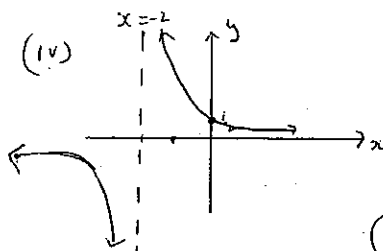
(2)



(2)



(2)



(2)

- (b) (i) function
(ii) function
(iii) not function
(iv) function

(4)

Question 1

(e) Domain: x all real
Range: $y > 0$

(2)

(ii) Domain: x all real
Range: $y \geq 0$

(2)

(iii) Domain: all real, $x \neq -2$
Range: all real, $y \neq 0$

(iv) Domain: $-2 \leq x \leq 2$
Range: $-2 \leq y \leq 2$

(d) Domain $8 - 2x > 0$
 $-2x > -8$
 $x < 4$

(2)

Question 2

(a) (i) $g(2) = 4 - 8$
 $= -4$

(2)

(ii) $g(1-w) = (1-w)^2 - 4(1-w)$
 $= 1 - 2w + w^2 - 4 + 4w$
 $= -3 + 2w + w^2$

(2)

(iii) $x^2 - 4x = -3$
 $x^2 - 4x + 3 = 0$
 $(x-3)(x-1) = 0$
 $x = 1 \text{ or } 3$

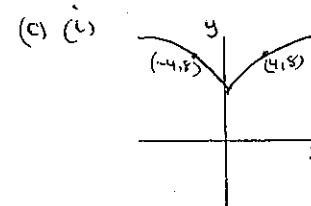
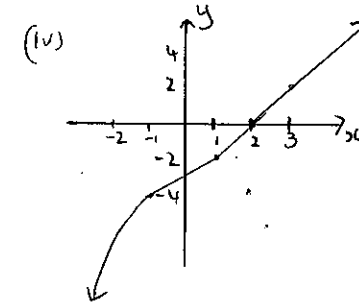
(2)

Question 2

(b) (i) $f(2) = 0$

(ii) $f(0) = -3$

(iii) $f(-1) = -4$



(ii) $f(4) + f(-4) = 16$

(d) $f(-x) = (-x)^3 - 4(-x)$
 $= -x^3 + 4x$
 $= -(x^3 - 4x)$
 $= -f(x)$

(2)

Question 3

(a) $\frac{(8-2) \times 180}{8} = 135^\circ$

(2)

Question 3

(b) $y + 105 + 110 + 105 = 360$
(angle sum of quad)
 $y = 40^\circ$

(2)

$x + 40 + 105 = 180$
(angle sum of triangle)

$x = 35^\circ$

(2)

(c) $\angle IHK = 90^\circ$ (straight angle)

3

$\angle IHK = \angle JKG = 70^\circ$
 $\therefore AF \parallel BE$ (corresponding angles are equal)

(d) $\frac{8}{3} = \frac{x}{5}$ (ratio of intercepts)

2

$x = \frac{40}{3}$

1

$AD = \frac{40}{3} + 5$

$= 18\frac{1}{3} \text{ m}$

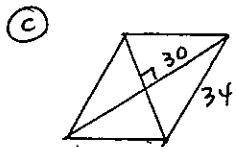
(4)

Question 4

(a) (i) $\frac{LC}{8} = \frac{9}{6}$ (ratio of intercepts)
 $LC = 12$ (2)

(ii) $AB^2 = 20^2 - 15^2$
 $AB = \sqrt{175}$ (2)

(b) LHS = $12^2 + 35^2$
 $= 144 + 1225$
 $= 1369$
 $= 37^2$
 $= RHS$
 $\therefore$ triangles right angled

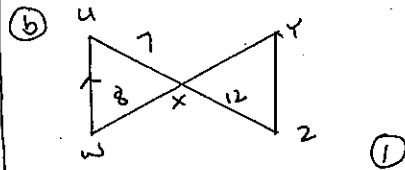


(i) $x^2 = 34^2 - 30^2$
 $= 256$
 $x = 16$
 $\therefore$ diagonal is 32cm

(ii) $\frac{1}{2} \times 32 \times 60 = 960 \text{ cm}^2$

Question 5

(a) $\hat{CPB} = 105^\circ$ (co-interior angles
 $AB \parallel CD$)
 $\hat{FDE} = 105^\circ$ (vertically opposite
angle)
 $\therefore \theta + 35 + 105 = 180$ (angle sum of $\triangle$)
 $\therefore \theta = 40^\circ$



(i) $\hat{YXZ} = \hat{UXW}$ (vertically
opposite angles)
 $\hat{WUX} = \hat{YZX}$ (alternate
angles, $uW \parallel YZ$)
 $\triangle uwx \parallel \triangle yxz$ (equilateral)

(3)

(ii) $\frac{24}{10} = \frac{12}{7}$ (proportional
sides of similar triangles)

$\therefore 24 = 17 \frac{1}{7}$

(c) Let $\hat{BEC} = x$

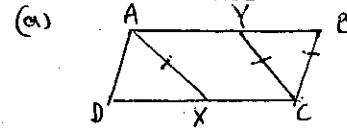
$\therefore \hat{ECF} = x$ (alternate
angles $BE \parallel CF$)

$\hat{EBC} = x$ (base angles
isosceles $\triangle EBC$, $EB = EC$)

$\hat{FCD} = x$ (corresponding
angles $BE \parallel CF$)

$\therefore \hat{ECF} = \hat{FCD} = x$
 $\therefore CF$ bisects $\hat{ECD}$

Question 6



(i) opposite angles
of parallelogram (1)

(ii) $BC = AD$ (opposite
sides of parallelogram)
 $AX = BC$ (given)

$\therefore AX = AD = BC$ (3)

(iv) $\hat{ADX} = \hat{BCY}$ (from part ii)

$\hat{AXD} = \hat{ADX}$ (angles
opposite equal sides of $\triangle$)

$\hat{ABY} = \hat{CBY}$ (angles opposite
equal sides of $\triangle$).

$\therefore \hat{AXD} = \hat{CXB} = \hat{CBY}$

$CY = AX$ (given)

$\therefore \triangle ADX \equiv \triangle CBY$ (AAS)

(4)

(vi) $DX = BY$ (corresponding sides
of congruent triangles)

$AB = DC$ (opposite sides of
parallelogram)

$AB = AY + YB$

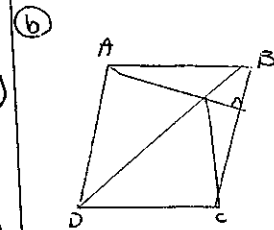
$DC = DX + XC$

$\therefore AY = XC = (AB - BY)$

$\therefore AX = YC$ (given)

$AXYC$ is a parallelogram
 $\therefore$ two pairs opposite sides equal

(4)



(i) In a rhombus diagonals
bisect angle through which
they pass.

(ii) $AD = DC$ (equal sides of rhombus)
 $\hat{ADB} = \hat{CDB}$ (from part i)

ED common

$\triangle AED \equiv \triangle CED$ (SAS) 4

(iii) $\hat{DAE} = 90^\circ$ (alternate angles
 $AD \parallel BC$) (1)

(iv) $\hat{CED} = 90^\circ$ (corresponding
angles of congruent $\triangle$'s).